

## KOZ'MODEM'YANSK, Russian Federation

WMO: 274790

Lat: 56.333N

Lon: 46.583E

Elev: 107

StdP: 100.04

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -26.3      | -23.0 | -29.2                         | 0.3 | -26.0 | -26.1 | 0.4 | -22.6 | 8.4                   | -6.6 | 7.9 | -7.4 | 1.6                   | 320  | 0.531 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.4                    | 29.6            | 20.4 | 27.9 | 20.0 | 26.3 | 19.2 | 21.8                | 27.5 | 20.9 | 26.3 | 20.0 | 25.0 | 2.2                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.8                            | 14.7 | 24.7 | 18.9 | 13.9 | 23.8 | 18.1 | 13.2 | 23.0 | 64.0          | 27.5 | 60.8 | 26.2 | 57.8 | 25.0 | 26.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.5               | 6.7  | 6.0 | -28.7                      | 31.7 | 3.7                | 1.9 | -31.3                                              | 33.1 | -33.5      | 34.2 | -35.5      | 35.3 | -38.2      | 36.7 |    |    |
|                   |      |     | -28.9                      | 23.1 | 3.6                | 1.3 | -31.5                                              | 24.0 | -33.6      | 24.8 | -35.6      | 25.5 | -38.3      | 26.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 5.3         | -9.7  | -9.1  | -2.9  | 5.9   | 13.8  | 17.9  | 20.4  | 18.2  | 12.4  | 5.3   | -2.0  | -7.3  |
|                                                                   | DBStd       | 11.97       | 7.02  | 6.86  | 4.53  | 5.10  | 5.16  | 4.37  | 3.68  | 4.09  | 4.29  | 4.43  | 5.47  | 6.74  |
|                                                                   | HDD10.0     | 2788        | 612   | 535   | 401   | 144   | 22    | 0     | 0     | 23    | 156   | 359   | 535   |       |
|                                                                   | HDD18.3     | 4949        | 870   | 768   | 659   | 374   | 157   | 59    | 21    | 52    | 183   | 404   | 609   | 793   |
|                                                                   | CDD10.0     | 1086        | 0     | 0     | 0     | 20    | 141   | 240   | 323   | 256   | 96    | 11    | 0     | 0     |
|                                                                   | CDD18.3     | 206         | 0     | 0     | 0     | 0     | 17    | 48    | 86    | 49    | 6     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 1259        | 0     | 0     | 0     | 1     | 99    | 304   | 547   | 291   | 17    | 0     | 0     | 0     |
|                                                                   | CDH26.7     | 303         | 0     | 0     | 0     | 0     | 16    | 71    | 135   | 80    | 2     | 0     | 0     | 0     |
| Wind                                                              | WSAvg       | 2.9         | 3.1   | 3.1   | 3.1   | 2.8   | 2.8   | 2.6   | 2.2   | 2.4   | 2.7   | 3.2   | 3.3   | 3.3   |
| Precipitation                                                     | PrecAvg     | 604         | 43    | 39    | 38    | 34    | 41    | 69    | 71    | 58    | 57    | 53    | 52    | 49    |
|                                                                   | PrecMax     | 750         | 93    | 74    | 71    | 82    | 67    | 164   | 146   | 121   | 115   | 104   | 88    | 89    |
|                                                                   | PrecMin     | 431         | 18    | 16    | 6     | 8     | 14    | 14    | 12    | 0     | 12    | 3     | 16    | 17    |
|                                                                   | PrecStd     | 71          | 16    | 16    | 17    | 20    | 15    | 39    | 35    | 33    | 29    | 28    | 18    | 16    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 2.5   | 2.8   | 9.1   | 21.8  | 28.7  | 31.2  | 32.7  | 32.4  | 25.4  | 18.2  | 10.7  | 5.5   |
|                                                                   |             | MCWB        | 1.5   | 1.2   | 5.0   | 12.9  | 18.5  | 20.7  | 21.0  | 21.1  | 16.9  | 12.5  | 8.8   | 4.6   |
|                                                                   | 2%          | DB          | 1.4   | 1.9   | 5.9   | 18.7  | 26.0  | 28.6  | 29.7  | 28.8  | 22.8  | 15.2  | 7.8   | 3.1   |
|                                                                   |             | MCWB        | 0.7   | 0.8   | 2.6   | 10.9  | 17.0  | 20.0  | 20.7  | 20.5  | 16.1  | 11.1  | 6.5   | 2.3   |
|                                                                   | 5%          | DB          | 0.7   | 1.1   | 4.0   | 16.3  | 23.8  | 26.8  | 27.9  | 26.2  | 20.8  | 13.1  | 6.0   | 1.8   |
|                                                                   |             | MCWB        | 0.0   | 0.3   | 1.5   | 9.9   | 15.8  | 19.3  | 20.5  | 19.4  | 15.2  | 10.1  | 4.9   | 1.0   |
|                                                                   | 10%         | DB          | -0.4  | 0.1   | 2.7   | 13.9  | 21.6  | 24.9  | 26.4  | 24.4  | 18.7  | 11.4  | 4.1   | 0.9   |
|                                                                   |             | MCWB        | -1.0  | -0.6  | 1.0   | 8.6   | 14.5  | 18.1  | 19.7  | 18.6  | 14.2  | 9.1   | 3.2   | 0.2   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 1.7   | 1.8   | 5.6   | 13.4  | 19.6  | 22.4  | 23.2  | 22.5  | 18.2  | 13.4  | 8.8   | 4.7   |
|                                                                   |             | MCDB        | 2.1   | 2.3   | 8.9   | 19.7  | 27.1  | 28.5  | 28.6  | 28.8  | 22.9  | 16.2  | 10.0  | 5.2   |
|                                                                   | 2%          | WB          | 0.9   | 1.0   | 3.1   | 11.8  | 17.9  | 21.1  | 21.9  | 21.4  | 16.8  | 11.7  | 6.8   | 2.4   |
|                                                                   |             | MCDB        | 1.3   | 1.5   | 5.1   | 16.9  | 24.7  | 26.7  | 27.4  | 27.1  | 21.4  | 14.2  | 7.8   | 2.9   |
|                                                                   | 5%          | WB          | 0.1   | 0.4   | 2.1   | 10.4  | 16.6  | 20.0  | 21.1  | 20.3  | 15.7  | 10.4  | 5.0   | 1.2   |
|                                                                   |             | MCDB        | 0.7   | 1.0   | 3.6   | 15.8  | 22.4  | 25.1  | 26.4  | 25.2  | 19.6  | 12.6  | 5.8   | 1.6   |
|                                                                   | 10%         | WB          | -1.0  | -0.6  | 1.1   | 8.8   | 15.4  | 18.9  | 20.3  | 19.2  | 14.6  | 9.3   | 3.3   | 0.3   |
|                                                                   |             | MCDB        | -0.4  | 0.2   | 2.4   | 13.6  | 20.7  | 24.1  | 25.3  | 23.4  | 18.1  | 11.2  | 4.0   | 0.8   |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.1   | 5.8   | 5.9   | 7.3   | 8.2   | 7.6   | 7.4   | 7.2   | 6.1   | 4.5   | 3.7   | 4.5   |
|                                                                   |             | MCDBR       | 4.0   | 4.3   | 6.4   | 10.5  | 10.2  | 9.4   | 9.1   | 9.2   | 8.8   | 6.8   | 4.8   | 3.7   |
|                                                                   | 5% WB       | MCWBR       | 3.9   | 4.1   | 4.5   | 5.5   | 4.9   | 4.4   | 4.0   | 4.1   | 4.5   | 4.4   | 4.3   | 3.4   |
|                                                                   |             | MCWBR       | 3.9   | 4.2   | 5.7   | 10.0  | 9.6   | 8.6   | 7.9   | 8.4   | 7.8   | 6.2   | 4.5   | 3.7   |
| Clear-Sky Solar Irradiance                                        | taub        |             | 0.286 | 0.318 | 0.355 | 0.404 | 0.381 | 0.376 | 0.404 | 0.425 | 0.371 | 0.350 | 0.313 | 0.282 |
|                                                                   |             | taud        | 2.078 | 2.059 | 2.031 | 2.143 | 2.356 | 2.406 | 2.335 | 2.253 | 2.389 | 2.338 | 2.214 | 2.059 |
|                                                                   | Ebn at Noon |             | 577   | 711   | 782   | 806   | 859   | 867   | 830   | 780   | 779   | 688   | 552   | 470   |
|                                                                   |             | Edh at Noon | 61    | 94    | 125   | 131   | 113   | 109   | 115   | 117   | 89    | 72    | 55    | 49    |
| All-Sky Solar Radiation                                           | RadAvg      |             | 0.57  | 1.43  | 2.92  | 4.07  | 5.44  | 5.80  | 5.71  | 4.44  | 2.69  | 1.23  | 0.48  | 0.31  |
|                                                                   | RadStd      |             | 0.09  | 0.28  | 0.33  | 0.41  | 0.37  | 0.46  | 0.51  | 0.44  | 0.33  | 0.19  | 0.08  | 0.08  |

## Historical Trends

|                        |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                    | (e)   | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only           | +0.61 | N/A   | N/A     | N/A    | N/A     | N/A   | -152  | -197        | N/A     | N/A     |         |
| Regional (8 neighbors) | +0.56 | N/A   | N/A     | N/A    | N/A     | N/A   | -132  | -178        | N/A     | N/A     |         |

Nomenclature: See separate page